

Measuring the Effect of Human Personality on LLM Token Consumption

Michael Anderson - HermanScience

May 5, 2026

Table of Contents

- 1. ABSTRACT3
- 2. RESEARCH QUESTION AND TEST OBJECTIVE.....3
- 3. EXPERIMENTAL HYPOTHESIS3
- 4. SYSTEM UNDER TEST4
- 5. EXPERIMENTAL DESIGN4
 - 5.1 INDEPENDENT VARIABLE4
 - 5.2 CONTROLLED VARIABLES.....4
 - 5.3 DEPENDENT VARIABLES.....4
 - 5.4 PERSONALITY CONDITIONS.....5
 - 5.5 RENDERED PROMPT STRUCTURE.....5
- 6. VALIDITY RATIONALE.....6
 - 6.1 INTERNAL VALIDITY.....6
 - 6.2 CONSTRUCT VALIDITY.....6
 - 6.3 EXTERNAL VALIDITY6
 - 6.4 MEASUREMENT VALIDITY6
- 7. EXECUTION WORKFLOW7
- 8. DATA CAPTURE AND AUDITABILITY7
- 9. ANALYSIS PLAN7
- 10. STATISTICAL AND ENGINEERING INTERPRETATION GUIDANCE8
- 11. KNOWN LIMITATIONS8
- 12. ACCEPTANCE CRITERIA FOR A VALID EXPERIMENT RUN.....8
- 13. FINDINGS SUMMARY9
 - 13.1 DATASET ANALYZED9
 - 13.2 PRIMARY FINDING: PERSONALITY-DRIVEN PROMPT STYLE AFFECTS TOKEN UTILIZATION.....9
 - 13.3 TOKEN USE AND RESPONSE QUALITY10
 - 13.4 TOKEN-EFFICIENCY AND RECOVERABLE SAVINGS ANALYSIS.....10
 - 13.5 VARIATION BY TASK TYPE.....10
 - 13.6 VARIATION BY PERSONALITY TYPE.....11
 - 13.7 OPTIMIZATION RULES AND ESTIMATED SAVINGS11
 - 13.8 STATISTICAL VARIATION AND CONFIDENCE.....12
 - 13.9 DATA-QUALITY NOTES AND EXCLUSIONS.....12
- 14. EXECUTIVE SUMMARY.....13
- 15. CONCLUSION14

1. Abstract

This report documents a second set of controlled LLM Prompt Eval v2 experiments designed to test whether human personality-driven prompting behavior affects LLM token consumption. The prior response-quality experiment tested whether personality-authored request styles affected output quality when protected task content was held constant. This experiment uses the same prompt-authorship structure, task families, personality conditions, model roles, execution workflow, and scalar evaluator scoring approach, but adds token utilization as the primary measurement dimension.

The experiment analyzed 21 completed runs across seven task types, with three experiments per task type. Each experiment included one neutral condition and nine personality-authored prompt conditions. Each condition was executed with repeated runs and captured generation input tokens, generation output tokens, generation total tokens, evaluator score, and generation tokens per score point. Pairwise comparison was disabled for this round and is not used in the findings.

The central finding is that personality-authored prompt style materially affects token utilization. The data also shows that higher token consumption does not reliably produce better output quality. Across all personality-authored variants, the correlation between scalar quality score and generation total tokens was negative, while the relationship between score and tokens per score point was more strongly negative. This indicates that better responses were often more token-efficient, not more token-expensive.

2. Research Question and Test Objective

The central research question is: when task content and output structure are held constant, does the personality-driven authorship style of the prompt materially affect the number of tokens used by the LLM to produce a response?

The test objective is to isolate prompt authorship as the manipulated condition and measure its effect on token utilization, response quality, and token efficiency. The experiment is not designed to test whether one task is easier than another, whether one output contract is better than another, or whether one personality type is better than another. It is designed to test whether different human prompting styles cause measurable differences in token consumption and whether those differences can be optimized without sacrificing response quality.

3. Experimental Hypothesis

Primary hypothesis: Personality-authored prompt variants will produce measurable differences in LLM token utilization even when the task objective, context, output contract, and scope boundaries are preserved across conditions.

Null hypothesis: When the task objective, context, output contract, and scope boundaries are preserved, personality-authored request wording will not produce meaningful differences in token utilization beyond normal model variance.

Secondary hypothesis: Token utilization will not have a simple positive relationship with response quality. Some prompting styles may cause the model to use more tokens without improving the answer, while other styles may guide the model toward higher-quality answers with fewer tokens per score point.

4. System Under Test

The LLM Prompt Eval v2 system is a local-first experiment harness for testing how prompt authorship affects LLM behavior. The current experiment uses the same system structure documented in the original response-quality report, with token capture added as a primary analysis dimension.

Component	Test-engineering function
Review UI / Builder	Guided experiment drafting, model-role selection, protected field entry, rendered prompt preview, and human approval before execution.
API service	Experiment builder endpoints, capability bootstrap, draft approval, execution metadata, result retrieval, and comparison surfaces.
Core package	Shared database models, schemas, settings, and validation logic.
Runner	Claims approved pending runs, executes model calls, captures outputs, records token usage, and processes batches locally.
Model under test	The generation model that receives each rendered prompt and produces the response being evaluated.
Evaluator model	A separate model role used to score outputs with scalar quality measurements.
Prompt transformer model	Role used during draft generation to support personality-authored request variants.
Database	Stores experiment definitions, runs, raw outputs, scalar summaries, token metrics, and execution status.

5. Experimental Design

5.1 Independent Variable

The independent variable is prompt authorship style as represented by personality-authored request wording. Each condition stores a `prompt_authorship_variant` and `rendered_prompt`. The variant may change the request wording, task framing, implied priorities, and social stance toward the model. The protected task content remains constant.

5.2 Controlled Variables

The design controls the task elements that would otherwise confound the experiment. The following fields are protected and must remain constant across variants: task objective, task context or source material, output contract, and scope boundaries. The rendered prompt format enforces this separation by placing authored request wording above the protected task fields.

5.3 Dependent Variables

The primary dependent variable for this report is generation token utilization. The secondary dependent variables are scalar response quality and token efficiency, calculated as generation tokens per scalar score point.

Measure	Purpose	Interpretation
Generation input tokens	Measures the prompt-side token cost sent to the model under test.	Used to confirm that protected task content and authorship variants create different prompt loads.
Generation output tokens	Measures the response-side token cost produced by the model.	Used to understand whether prompt style causes longer or shorter answers.
Generation total tokens	Primary token-consumption measure for the model under test.	Used to compare token usage across personality variants.
Scalar quality score	Summarizes output quality using the	Used to determine whether lower token use

	evaluator model.	preserves, improves, or reduces response quality.
Generation tokens per score point	Measures token efficiency.	Lower values indicate more quality produced per generation token.
Grouped results by personality and task	Aggregates outcomes by modeled personality category and task family.	Used to identify recurring token-efficiency patterns.
Raw response JSON	Preserves execution and scoring artifacts.	Used for audit, troubleshooting, and deeper offline analysis.

5.4 Personality Conditions

The experiment model supports one neutral condition and nine named personality conditions. The current findings focus on the nine personality-authored conditions. Neutral was retained for reference, but excluded from personality best-versus-worst spread calculations and optimization-rule estimates.

Condition family	Conditions
Baseline	neutral
Personality-authored variants	Reformer, Helper, Achiever, Individualist, Investigator, Loyalist, Enthusiast, Challenger, Peacemaker
Trait metadata used for authorship	structure, directness, exploration, warmth, constraint, risk_focus

5.5 Rendered Prompt Structure

Every rendered prompt follows the same structure used in the response-quality experiment. This structure is a key validity control because it separates the manipulated authorship field from the protected task fields:

Request:

<personality-authored request wording>

Task objective:

<protected task objective>

Context:

<protected source/context material>

Output contract:

<protected output contract>

6. Validity Rationale

6.1 Internal Validity

The experiment is internally valid to the extent that observed token-consumption differences can be attributed to the prompt-authorship condition rather than uncontrolled changes in the task. The design supports internal validity through protected content preservation, structured rendered prompt format, role separation, human approval, and validation checks that reject prompts if protected content is dropped or mutated.

Threat to validity	Design control	Residual risk
Task content drift	Task objective, context, and output contract are stored separately and preserved in rendered prompts.	Semantic drift may still occur if authored request wording implies priorities that conflict with the output contract.
Prompt length contamination	Prompt variants may differ in wording length because this is part of natural authorship behavior.	Input-token differences are real measured effects, but output-token effects must be interpreted separately.
Output contract inconsistency	The output contract is protected and included in every rendered prompt.	Models may still interpret priority cues differently across authored request variants.
Evaluator availability mismatch	Capability bootstrap marks evaluator eligibility only when the runner supports the model in evaluation mode.	Evaluator model behavior remains a source of measurement noise.
Run-state ambiguity	Experiment-level status rolls up queued, running, completed, failed, and completed_with_failures states.	Partial failures must be excluded or separately flagged.

6.2 Construct Validity

The construct being tested is personality-influenced prompt authorship: the way a person frames a request, signals priorities, manages ambiguity, expresses constraints, and positions the relationship with the model. For this report, the construct is extended from output quality to token economics. The measured question is whether these authorship differences change how much work the model does and how efficiently that work produces useful output.

6.3 External Validity

External validity depends on the breadth of task types, model families, and prompt variants included in the final run plan. This experiment used seven task families and repeated experiments per task family. Results should still be treated as directional until repeated across additional models, additional prompt formats, and production usage contexts.

6.4 Measurement Validity

Token counts are direct execution measurements and are therefore stronger mechanical measurements than evaluator scores. However, token efficiency depends on both token count and scalar quality score. The evaluator model remains a measurement instrument, not ground truth. The strongest interpretation comes from analyzing token counts, quality scores, and tokens per score point together.

7. Execution Workflow

The operational workflow mirrors the prior response-quality experiment. The current workflow is:

1. Open the builder UI.
2. Bootstrap available model capabilities.
3. Enter experiment details, task objective, task context, output contract, personality types, and execution roles.
4. Generate the draft YAML.
5. Verify that the rendered prompt sample includes the full protected content.
6. Approve and queue the experiment.
7. Allow the runner to process queued runs.
8. Review run execution state, scalar summary scores, and token utilization metrics.

8. Data Capture and Auditability

The experiment captures both human-readable results and raw execution artifacts. For this report, the critical audit artifacts are the scalar evaluator scores and the generation token fields captured for each run and summarized at the variant level.

Artifact	Why it matters
Compiled YAML artifact	Provides a durable, reviewable definition of the experiment before execution.
Rendered prompt	Shows the exact prompt sent to the model under test.
Run status and counts	Supports execution traceability and failure triage.
Model response	Primary observed output from the model under test.
Generation token metrics	Primary token-consumption measurement for input, output, and total token use.
Evaluator result	Primary automated quality measurement.
Raw JSON	Audit trail for offline validation, debugging, and reproducibility checks.

9. Analysis Plan

The analysis was performed at four levels: aggregate token behavior, task-type sensitivity, personality-condition effect, and optimization potential. Pairwise comparisons were disabled for this round and were not used.

Analysis view	Question answered	Recommended output
Overall token effect	Do personality-authored prompts materially change token utilization?	Mean token usage, token range, best-worst spread, confidence interval.
Quality relationship	Do higher-token responses produce higher-quality answers?	Correlation between score and generation tokens; correlation between score and tokens per score point.
Task-type sensitivity	Which task types show the largest token-efficiency differences?	Grouped results by summarization, decision support, writing, explanation, recommendation, ideation, and risk analysis.
Personality-condition effect	Which personality-authored variants are more or less token efficient?	Ranked personality table with score, tokens, and tokens per score point.
Optimization estimate	What savings may be possible if prompts are transformed using personality and task information?	Quality-guarded rule set and estimated token savings by task and personality.
Failure analysis	Which runs failed or produced unusable outputs?	Failed-run list and exclusion rationale.

10. Statistical and Engineering Interpretation Guidance

The final interpretation should avoid treating token reduction as inherently good. The engineering objective is not to minimize tokens at all costs. The objective is to minimize unnecessary tokens while preserving or improving quality. For that reason, this report uses generation tokens per score point and quality-guarded optimization estimates rather than raw token minimization alone.

The report also distinguishes between maximum theoretical savings and practical savings. The maximum spread between the least and most token-efficient variants shows the size of the observed optimization space. The quality-guarded rule estimate is more conservative because it only recommends a lower-token prompt style when the selected style remains within 95% of the strongest average score for the same task family.

11. Known Limitations

Some providers and models do not support seeded deterministic behavior, which may introduce run-to-run variance.

Evaluator models are scalable measurement instruments but are not perfect substitutes for human expert review.

Token counts are mechanically captured, but token efficiency depends on scalar quality scores, which remain evaluator-derived.

The quality-guarded savings estimate uses the best observed task-aligned style within this data set; it should be validated on future experiments before being treated as a production savings guarantee.

This experiment analyzed one model-under-test configuration. Model-specific behavior may change the size and direction of the token effect.

Pairwise comparisons were disabled and are not part of this analysis.

12. Acceptance Criteria for a Valid Experiment Run

Criterion	Pass condition
Protected content preserved	Rendered prompt sample contains the complete task objective, task context, and output contract.
Condition isolation	Only request wording, framing, implied priorities, and social stance vary by personality condition.
Role selection verified	Model under test, evaluator model, and prompt transformer model are separately selected and capability-validated.
Human approval complete	Draft YAML and rendered prompt sample are reviewed before approval.
Execution complete	Experiment status is completed, or completed_with_failures with failures explicitly documented.
Scoring complete	Scalar score is available for all included runs.
Token capture complete	Generation input, output, total tokens, and tokens per score point are available for included runs.
Audit artifacts retained	Rendered prompt, model output, evaluator result, token metrics, and raw JSON remain available for inspection.

13. Findings Summary

13.1 Dataset Analyzed

The findings summarized in this section are based on experiments 198 through 218. The analyzed set includes 21 completed experiments across seven task types, with three experiments per task type. Each experiment includes one neutral baseline condition and nine personality-authored conditions. Each condition was executed with 10 repeated runs, producing 2,100 total runs and 1,890 personality-authored runs. Pairwise comparison was disabled and is not used in this report.

Dataset element	Count / scope
Experiments analyzed	21
Experiment ID range	198-218
Task types	7
Experiments per task type	3
Personality-authored variants	9
Neutral baseline	Included for reference; excluded from personality spread and savings calculations
Runs per condition	10
Total runs	2,100
Primary token measurement	Generation total tokens
Primary efficiency measurement	Generation tokens per score point
Quality measurement	Scalar evaluator score
Pairwise comparisons	Disabled; ignored for this report

13.2 Primary Finding: Personality-Driven Prompt Style Affects Token Utilization

Across the 21 completed experiments, the average observed token range between the highest-token and lowest-token personality-authored style within the same experiment was 48.2% of the highest-token style. The 95% confidence interval for this observed token-consumption spread was 40.9% to 55.5%. This indicates that personality-authored prompt style materially affects how many tokens the model uses to complete the same protected task.

Primary token finding	Result
Average observed token-consumption spread	48.2%
95% confidence interval for observed spread	40.9% to 55.5%
Average best-worst generation token delta	605
Average score spread within the same experiments	1.69 points
Average tokens-per-score-point efficiency spread	47.4%
95% confidence interval for efficiency spread	40.8% to 54.1%

The effect is large enough to matter operationally. In practical terms, two users can ask for the same underlying work, using the same protected task objective and output contract, and still cause the model to consume materially different numbers of tokens because their request framing leads the model into different response behaviors.

13.3 Token Use and Response Quality

The data does not support the assumption that more tokens reliably produce better answers. Across the personality-authored variants, the correlation between scalar quality score and generation total tokens was -0.338. The correlation between scalar quality score and generation tokens per score point was -0.639.

Relationship measured	Correlation	Interpretation
Quality score vs. generation total tokens	-0.338	Weak-to-moderate negative relationship. More tokens did not reliably produce better scores.
Quality score vs. generation tokens per score point	-0.639	Moderate-to-strong negative relationship. Higher-quality outputs were often more token efficient.

This is the most important product finding. Token optimization does not have to mean quality reduction. In this data set, many lower-token responses also scored better because the prompt formulation gave the model clearer constraints, better prioritization cues, or a more efficient path to the answer.

13.4 Token-Efficiency and Recoverable Savings Analysis

The report uses two savings views. The first is the observed efficiency spread: the difference between the best and worst tokens-per-score-point result within the same experiment. The second is a conservative quality-guarded optimization estimate: for each task family, select the lowest tokens-per-score-point style that remains within 95% of the best average quality score for that task family.

Savings view	Calculation	Interpretation
Observed token-consumption spread	(Highest generation tokens - lowest generation tokens) / highest generation tokens	Shows the maximum observed token-reduction space within each experiment.
Observed efficiency spread	(Worst tokens per score point - best tokens per score point) / worst tokens per score point	Shows the maximum observed improvement in quality-per-token efficiency.
Quality-guarded savings estimate	Savings versus the lowest-token-per-score style that remains within 95% of the best average task score	More conservative estimate for practical prompt-transformation rules.

The average observed efficiency spread was 47.4%, with a 95% confidence interval of 40.8% to 54.1%. The more conservative quality-guarded rule estimate produced an average potential savings of 12.8%, with a 95% confidence interval of 9.7% to 15.8%. This conservative estimate is the better planning assumption for production economics because it avoids recommending lower-token prompts that materially reduce quality.

13.5 Variation by Task Type

Token sensitivity varied by task type. The largest observed token-consumption spreads appeared in explanation, recommendation, risk analysis, decision support, writing, and ideation. Summarization showed the smallest spread, which is consistent with the bounded nature of the task.

Task type	Avg. token spread	Avg. efficiency spread	Avg. token delta	Avg. score spread
Explanation	65.0%	68.2%	898	2.02
Recommendation	57.7%	53.4%	697	1.84
Risk Analysis	55.2%	51.8%	758	1.65
Decision Support	47.4%	47.5%	620	1.71
Writing	47.0%	34.3%	469	1.69
Ideation	46.3%	52.7%	657	1.57
Summarization	18.7%	24.0%	135	1.33

The task-level pattern differs from a simple quality-lift pattern. Some judgment-heavy tasks used many tokens, but those extra tokens did not consistently translate into better quality. This suggests that prompt transformation should not only make prompts more complete. It should also prevent personality-driven overproduction when the task can be answered with fewer tokens.

13.6 Variation by Personality Type

Personality-level results show meaningful differences in average token use and average token efficiency. The table below ranks personality-authored styles by generation tokens per score point. Lower values indicate better token efficiency.

Personality style	Avg. score	Avg. gen tokens	Avg. input tokens	Avg. output tokens	Tokens per score point
Challenger	5.92	626	193	156	108.35
Investigator	6.41	789	194	317	127.53
Peacemaker	6.76	929	194	453	143.16
Helper	6.81	999	197	526	151.75
Enthusiast	6.46	962	195	487	152.65
Achiever	6.53	996	193	497	156.92
Loyalist	6.47	1,075	195	574	173.23
Reformer	6.46	1,072	197	569	173.87
Individualist	6.40	1,161	195	606	191.27

The Challenger style was the most token efficient on a raw tokens-per-score basis, but it also had the lowest average quality score. Individualist, Reformer, and Loyalist used the most tokens per score point. This shows why raw token minimization is not enough. A production prompt transformer should use personality and task together, then apply a quality guard so that savings do not come from simply shortening answers that score worse.

13.7 Optimization Rules and Estimated Savings

To estimate practical savings, the analysis selected one task-level optimization rule for each task family. The selected rule is the personality-authored style with the lowest average generation tokens per score point while remaining within 95% of the best average score for that task family. This is a conservative proxy for what a prompt transformer might do: preserve the task intent, render the prompt in a more efficient task-aligned style, and avoid material quality loss.

Task type	Quality-guarded efficient style	Avg. score	Best task avg. score	Avg. tokens per score point
Decision Support	Investigator	5.53	5.65	166.68
Explanation	Peacemaker	7.93	7.93	128.14
Ideation	Helper	6.56	6.56	198.70
Recommendation	Peacemaker	6.95	7.29	96.72
Risk Analysis	Achiever	6.34	6.34	167.66
Summarization	Investigator	7.54	7.71	80.40
Writing	Peacemaker	7.10	7.33	100.80

Using these task-level rules, the average estimated savings across personality-task combinations was 12.8%. The 95% confidence interval was 9.7% to 15.8%. This is the report's recommended planning estimate for near-term savings: roughly 10% to 16% generation-token savings while preserving near-best observed quality. The larger observed efficiency spread of about 47.4% represents the upper bound of the optimization space, not the conservative production estimate.

Task type	Avg. quality-guarded savings estimate
Decision Support	18.4%
Recommendation	16.9%
Explanation	16.3%
Risk Analysis	15.2%
Writing	8.8%
Summarization	8.0%
Ideation	5.7%

Source personality style	Avg. quality-guarded savings opportunity
Individualist	27.3%
Loyalist	21.5%
Reformer	21.1%
Achiever	13.5%
Enthusiast	13.0%
Helper	11.0%
Peacemaker	5.9%
Investigator	1.1%
Challenger	0.4%

13.8 Statistical Variation and Confidence

The overall observed token-consumption spread across 21 experiments was 48.2%, with a standard deviation of 17.0% and a standard error of 3.7%. The estimated 95% confidence interval was 40.9% to 55.5%. The observed tokens-per-score-point efficiency spread was 47.4%, with a 95% confidence interval of 40.8% to 54.1%.

Statistical measure	Observed token-consumption spread	Tokens-per-score-point efficiency spread	Quality-guarded rule estimate
Mean	48.2%	47.4%	12.8%
Standard deviation	17.0%	15.6%	12.3%
Standard error	3.7%	3.4%	1.5%
95% confidence interval	40.9% to 55.5%	40.8% to 54.1%	9.7% to 15.8%
Sample basis	21 experiments	21 experiments	63 task-personality combinations

The overall token effect has moderate confidence. The task-level ranking should be treated as directional because each task family has three experiments. The quality-guarded savings estimate is more conservative and more useful for product planning, but it should still be validated against additional models and production prompt traffic before being used as a guaranteed financial claim.

13.9 Data-Quality Notes and Exclusions

The neutral baseline was retained for reference but excluded from personality-authored best-versus-worst spread and savings calculations. This keeps the analysis aligned to the question of personality-driven prompt behavior. Pairwise comparisons were disabled in this experiment set and are not used. All included experiments were completed and contained summary-level scalar scores and token metrics.

Generation token metrics were used for the primary findings. Evaluation tokens were captured but not treated as the primary economic measure because evaluator-token usage is part of the test harness, not the user-facing model-under-test interaction. Production economics should be based primarily on the generation model tokens consumed in serving the user response.

14. Executive Summary

The results from experiments 198 through 218 provide empirical support for the hypothesis that personality-driven prompt authorship affects LLM token consumption when task content, output requirements, model role, evaluator role, and execution structure are controlled.

The most important finding is that token usage and output quality are not positively coupled. More tokens did not reliably produce better answers. Across the personality-authored variants, the correlation between quality score and generation total tokens was negative. The correlation between quality score and generation tokens per score point was more strongly negative. In practical terms, the model often produced better answers when the prompt style guided it to use tokens more efficiently.

Executive finding	Result
Experiments analyzed	21
Task types represented	7
Personality-authored runs analyzed	1,890
Average observed token-consumption spread	48.2%
95% confidence interval for token spread	40.9% to 55.5%
Average observed tokens-per-score efficiency spread	47.4%
Quality score vs. generation-token correlation	-0.338
Quality score vs. tokens-per-score correlation	-0.639
Conservative quality-guarded savings estimate	12.8%
95% confidence interval for conservative savings	9.7% to 15.8%

The observed optimization space is large. Within the same experiment, the average spread between the highest-token and lowest-token personality-authored style was approximately 48%. The average spread between the worst and best tokens-per-score-point result was approximately 47%. These are not recommended savings claims. They show the size of the measured inefficiency created by prompt-style variation.

The more conservative planning estimate is the quality-guarded rule estimate. Using a rule that selects the most token-efficient task style while staying within 95% of the best average task score, the estimated generation-token savings averaged approximately 12.8%, with a 95% confidence interval of approximately 9.7% to 15.8%. This is a more credible near-term estimate for what may be achievable by a prompt transformer that knows the user personality profile and the task type.

The results therefore support the product thesis: if the system knows the personality of the prompter and can infer the task type, it should be possible to transform the prompt into a task-aligned rendering that preserves intent, improves or protects response quality, and reduces avoidable token consumption. The value is not merely better prompting. It is better quality per token.

15. Conclusion

The LLM Prompt Eval v2 harness provides a credible experimental structure for isolating the effect of human personality-driven prompt authorship on LLM token consumption. The key engineering strength remains the separation of mutable authored request wording from protected task content. This enables a cleaner causal interpretation than experiments that change the whole prompt or merely instruct the model to answer in a different tone.

The new token-utilization results extend the prior quality findings. Personality-driven prompting behavior does not only affect response quality. It also changes how many tokens the model uses and how efficiently those tokens produce quality. The data indicates that token optimization based on personality and task alignment is plausible, but it should be implemented with a quality guard rather than as simple token minimization.

For product planning, the recommended working estimate from this experiment set is approximately 10% to 16% generation-token savings under a quality-guarded prompt-transformation strategy. Larger savings may be possible in specific task and personality combinations, but those estimates should be validated with additional experiments, additional models, and production traffic before being represented as general savings claims.